

# Practice Quiz 04C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

*Families of sets, union, intersection, and inclusion.*

1. Let  $a, b, c$  be three distinct objects, and let

$$\mathcal{F} = \{\{a\}, \{a, b\}, \{a, c\}\}.$$

- (a) Compute  $\bigcup \mathcal{F}$  and  $\bigcap \mathcal{F}$ .

**Solution** The elements belonging to at least one member of  $\mathcal{F}$  are  $a, b, c$ , so  $\bigcup \mathcal{F} = \{a, b, c\}$ . Only  $a$  belongs to every member, so  $\bigcap \mathcal{F} = \{a\}$ . ■

- (b) Decide whether  $\{a\} \in \mathcal{F}$  and whether  $\{\{b\}\} \subseteq \mathcal{F}$ . Explain each answer using the elements listed in  $\mathcal{F}$ .

**Solution** The first statement is true because  $\{a\}$  is one of the three listed elements of  $\mathcal{F}$ . The second is false: the sole element of  $\{\{b\}\}$  is  $\{b\}$ , and  $\{b\}$  is not any of  $\{a\}$ ,  $\{a, b\}$ , or  $\{a, c\}$ , since  $a, b, c$  are distinct. Thus not every element of  $\{\{b\}\}$  belongs to  $\mathcal{F}$ . ■